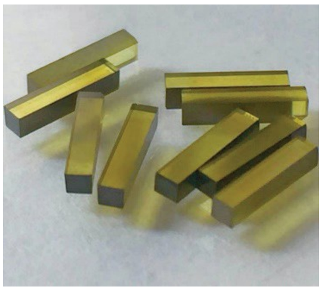
MCD – Single Crystal

Developed using HPHT technology (high pressure, high temperature), the manufacture of synthetic monocrystalline diamond is achieved by combining a mixture of carbon and transitional metals at very high pressures and temperatures.
Properties: Excellent thermal conductivity and good wear resistance & repeatability of chemical, physical and thermal characteristics.
Shapes available: Topped octahedron shapes, rhombic-dodecahedron shape or simply plate like crystals.
Color: Yellow
Qualities: The rough synthetic stones are carefully sorted into various qualities and grades. Each grade reflects either our grade 1 flawless crystal or grade 2 & 3 with various levels of internal inclusions

Chemical and physical properties

Hardness	10,000 kg/mm2
Density	3.51 g/cm3
Young´s modulus	1.22 Gpa
Index of refraction at 650 nm	2.41
Thermal conductivity	1200 W/mK
Optical transmissivity at 650 nm	65%
Wear resistance	up to 15% higher than natural diamond
Complete impurity absorption	450-500 nm

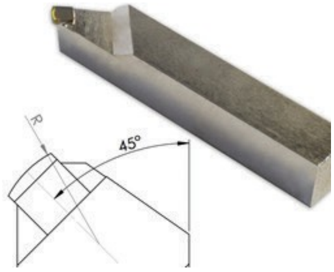


Above are Typical MCD products from our range.

Dimensions available are all sizes up to 8mm x 8 mm x 1.60-1.70
mm Our MCD is supplied in 100 orientation (4-point diamond)
The required dimensions always depend on the customer’s application. In general EID delivers diamonds with diameters up to 5 mm. The weight is shown in carats.

EID Ltd, EID House, 12 St Cross Street, London EC1N 8UB. England.
Tel: +-44-207-405-6594. Fax: +-44-207-831-0372. Website: www.eid-ltd.com. Email- lee@eid-ltd.com , phil@eid-ltd.com

Types of Mono Crystal Diamond Cutting Tools



Above are some typical mono crystal cutting tools.

Mono crystal diamond cutting tools are suitable for processing many materials:

Including gold, silver, copper, copper alloys, aluminum, aluminum alloys, beryllium alloys, oxygen-free copper (OFC), electroless nickel, graphite, glass, plastic, ceramics, un-sintered cemented carbide, Various fiber and particle reinforced composite materials, rubber and various wear-resistant wood (especially solid wood and plywood, MDF and other composite materials)

Advantages of MCD Cutting Tools

• High hardness and wear resistance

• High efficiency and low cost

• High surface finish can be obtained on your workpiece

Mono-crystal diamond tools are often used for machining high - precision parts with mirror finish. The mono-crystal diamond tools can achieve excellent cutting edge by grinding. The surface finish can reach 0.025 micron or higher.

Applications of Mono Crystal Diamond Tools

Optical industry: Optical lenses, (aspheric) spherical lenses, optical glass, optical molds, mirrors, etc.

Printing industry: roller mold.

Automotive industry: night driving optical system, projection lights, aluminum alloy wheels.

3D industry: mobile phone outer mirror/button mirror/torch outer mirror, computer hard disk substrate.

Electronic appliances: computer hard disk substrate.

Defense industry / aerospace: navigation gyro for missiles.

New materials: ceramics, engineering plastics.

Medical equipment: contact lenses, accelerator electron guns.

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